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[Test Procedure: Network Storm DoS Test]

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[Operator Information]

Operator	Position	Name	Signature

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1. Pre-Test Checklist

2. Test Contents & Procedure

Relevant Standard	Standard No.
	Article.
Purpose	The purpose of this test is to verify the system and hardware installation status before executing the Network Storm (DoS) test.
	Procedure of test Pass Fail

Test Procedure
Company Confidential

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1. Verify system installation status. - HiNAS system is installed on board - Firewall is installed and physically connected to HiNAS - L2 switch for HiNAS / BMS network is installed			Pass	Pass
2. Verify system health status. - HiNAS system boots normally without error - No abnormal alarm or fault message is displayed on HiNAS UI				
Comment / Recommended				

2.2 Network Storm (DoS) Test

Relevant Standard	IACS UR E27.
	Cyber Resilience of On-board Systems and Equipment
Purpose	The purpose of this test is to verify that the HiNAS system maintains normal operation when excessive network traffic (Network Storm / DoS condition) is generated, and that such abnormal traffic is properly detected and logged by the firewall.

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Procedure of test		Pass	Fail
1. Confirm that the HiNAS system is operating normally before the test. 2. Connect the test laptop to the L2 switch connected to the BMS network, and configure the test laptop IP address as the BMS IP. 3. Set the target IP address to the HiNAS system. 4. Execute the DoS (Network Storm) traffic generation script with the following configuration: - Protocol: UDP - Target IP: 10.0.100.20 (HiNAS) - Source IP: 10.0.50.12 (BMS) - Target Port: 5005 - Packet Size: 1024 bytes - Traffic Load: configured to generate approximately 300 Mbps of traffic 5. During the traffic generation, verify the following: - Firewall bandwidth usage increases and DoS-related detection is observed - HiNAS system continues normal operation without malfunction - No abnormal delay, freeze, or loss of display is observed on the HiNAS UI 6. Stop the DoS traffic after sufficient observation time. 7. Review firewall logs and HiNAS system logs to confirm that Network Storm / DoS events are properly recorded.		Pass	Pass
Comment / Recommended			